

Gosford High School

Year 11

2008

Preliminary Higher School Certificate

Mathematics Extension 1 Assessment Task 2

Time Allowed - 60 minutes
+ 5 minutes reading time

The three questions are of equal value

Remember to start each new question on a new page

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

GOSFORD HIGH SCHOOL – HALF YEARLY EXAMINATION 2008

MATHEMATICS EXTENSION 1 – YEAR 11

Time Allowed: 1 Hour + 5 minutes reading time

Question 1 (18 Marks)

Marks

- a) Write $(1 + \sqrt{5})^3$ in the form $a + b\sqrt{5}$ where a and b are integers. 3
- b) By expressing $\frac{4}{2 + \sqrt{5}} - \frac{1}{9 - 4\sqrt{5}}$ 3
in its simplest form, show that it is a rational number.
- c) Simplify $\frac{2^{n+3} - 2^{n-1}}{2^{2n+3} - 2^{2n-1}}$ showing all steps 3
- d) Solve $9^x - 28(3^x) + 27 = 0$ 3
- e) Solve and graph the solution on a number line
- (i) $\frac{4}{5 - x} \geq 1$ 3
- (i) $\frac{x + 1}{x^2 - 4} \geq 0$ 3

Question 2 (18 Marks) Begin a new sheet of paper

The points P and Q have coordinates (3, -2) and (1, 3) respectively.

- a) The line k has equation $4x + 5y - 2 = 0$. Verify that P lies on k . 1
- b) The line l through Q has gradient $\frac{1}{3}$. Show that the equation of l is 2
 $x - 3y + 8 = 0$.
- c) Find the point of intersection R of k and l . 2
- d) Find the perpendicular distance of P from l . 3
- e) Hence find the area of the triangle PQR 3
- f) Find the point T which divides PQ externally in the ratio 1:2 3
- g) Find the equation of the line concurrent with 4
 $4x + 5y - 2 = 0$ and $x - y + 1 = 0$
which is perpendicular to $3x + 2y + 1 = 0$.

Question 3 (18 Marks) Begin a new sheet of paper**Marks**a) Solve for $0 \leq \theta \leq 360^\circ$

(i) $3 \sin \theta + 2 \cos \theta = 0$ (Nearest minute)

2

(ii) $\cos 2\theta = \cos \theta$ (Exact Answers)

3

b) A yacht sails 16 Km from a port P on a bearing of 102° to a buoy Q. From Q, it then sails 23 Km on a bearing of 207° to another buoy R.

(i) Draw a large diagram showing this information.

1

(ii) Prove that $\angle PQR$ is 75°

1

(iii) Find how far the yacht is from P. (Answer to 2 decimal places)

2

(iv) On what bearing must it sail to return to P? (Nearest minute)

3

c) State the expansion of $\cos(\alpha + \beta)$ and hence find the exact value of $\cos 105^\circ$

3

d) Prove showing all steps:

$$\frac{\cos(180^\circ - A)}{\sin(180^\circ + A)} - \frac{\sin(-A)}{\sin(90^\circ - A)} = \sec A \operatorname{cosec} A$$

3

Year 11 Ext 1 H/Y Solutions

Question 1

$$\begin{aligned} a) (1+\sqrt{5})^3 &= (1+\sqrt{5})(1+2\sqrt{5}+5) \\ &= (1+\sqrt{5})(6+2\sqrt{5}) \\ &= 6+2\sqrt{5}+6\sqrt{5}+10 \\ &= 16+8\sqrt{5} \\ \therefore a=16, \quad b=8 \end{aligned}$$

$$\begin{aligned} b) \frac{4}{2+\sqrt{5}} \times \frac{2-\sqrt{5}}{2-\sqrt{5}} &= \frac{8-4\sqrt{5}}{4-5} \\ \frac{1}{9-4\sqrt{5}} \times \frac{9+4\sqrt{5}}{9+4\sqrt{5}} &= \frac{9+4\sqrt{5}}{81-80} \\ \therefore -(8-4\sqrt{5}) - (9+4\sqrt{5}) \\ &= -8+4\sqrt{5}-9-4\sqrt{5} \\ &= -17 \text{ which is rational} \end{aligned}$$

$$\begin{aligned} c) \frac{2^{n+3} - 2^{n-1}}{2^{2n+3} - 2^{2n-1}} \\ \frac{2^{n-1} \{2^4 - 1\}}{2^{2n-1} \{2^4 - 1\}} &= 2^{n-1-2n+1} \\ &= 2^{-n} = \frac{1}{2^n} \end{aligned}$$

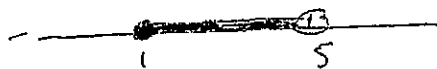
$$\begin{aligned} d) 9^x - 28(3^x) + 27 &= 0 \\ 3^{2x} - 28(3^x) + 27 &= 0 \\ \text{Put } u &= 3^x \\ u^2 - 28u + 27 &= 0 \\ (u-27)(u-1) &= 0 \\ u &= 27 \text{ or } 1 \\ 3^x &= 27 \text{ or } 3^x = 1 \\ x &= 3 \text{ or } x = 0 \end{aligned}$$

$$e) i) \frac{4}{5-x} \geq 1$$

Critical Points: $x=5$

$$\text{And } 4 = 5-x$$

$$x=1$$



$$\text{Test } x=0 \quad \frac{4}{5} \not\geq 1$$

$$1 \leq x < 5$$

$$ii) \frac{x+1}{x^2-4} \geq 0$$

$$x \neq \pm 2$$

$$\begin{aligned} \text{Solve } x+1 &= 0 \\ x &= -1 \end{aligned}$$



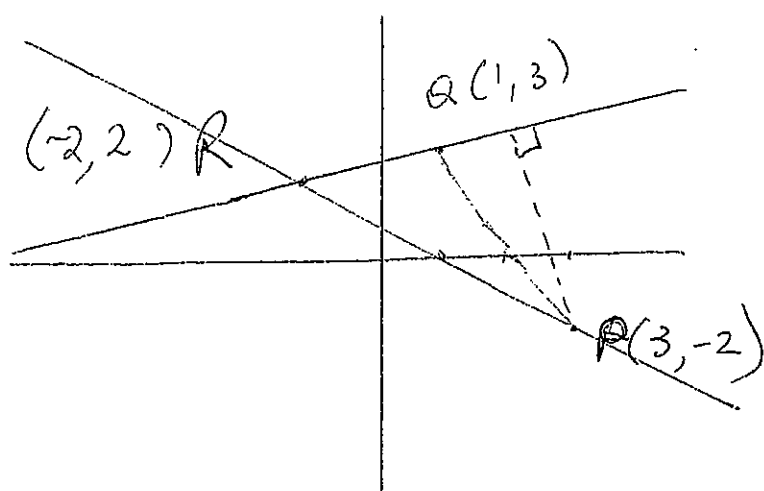
$$\text{Test } x=0 \quad \frac{1}{-4} \not\geq 0$$

$$\text{Test } x=3 \quad \frac{4}{5} \geq 0 \quad \checkmark$$

$$\text{Test } x=-\frac{1}{2} \quad \frac{-\frac{1}{2}}{-\frac{9}{4}} \geq 0 \quad \checkmark$$

$$-2 < x < 2 \text{ or } x > 2$$

Question 2



a) $P(3, -2)$ $4x + 5y - 2 = 0$
 $LHS = 4 \times 3 + 5 \times -2 - 2$
 $= 12 - 10 - 2$
 $= 0$
 $\therefore P$ lies on l

b) $m = \frac{1}{3}$ $Q(1, 3)$
 $y - 3 = \frac{1}{3}(x - 1)$
 $3y - 9 = x - 1$
 $x - 3y + 8 = 0$ is l

c) $4x + 5y = 2$ — (1)
 $x - 3y = -8$ — (2)
 $(2) \times -4$
 $-4x + 12y = 32$ — (3)
 $(1) + (3)$
 $17y = 34$
 $y = 2$
 Sub into (2)
 $x = 6 - 8$
 $x = -2$
 R is $(-2, 2)$

d) $d = \left| \frac{1 \times 3 - 3 \times -2 + 8}{\sqrt{1^2 + 3^2}} \right|$
 $d = \frac{17}{\sqrt{10}}$

e) QR :
 $d^2 = 3^2 + 1^2$
 $d = \sqrt{10}$
 $\therefore \text{Area } \triangle PQR = \frac{1}{2}bh$
 $= \frac{1}{2} \times \sqrt{10} \times \frac{17}{\sqrt{10}}$
 $A = \frac{17}{2} \text{ u}^2$

f) $\frac{k}{2} = \frac{-1}{2}$
 $x = \frac{-1 \times 1 + 2 \times 3}{-1 + 2}$ $y = \frac{-1 \times 3 + 2 \times -2}{-1 + 2}$
 $x = 5$ $y = -7$
 T is $(5, -7)$

g) $(4x + 5y - 2) + k(x - y + 1) = 0$
 $m_1 = -\frac{(4+k)}{5-k}$ $m_2 = -\frac{3}{2}$
 $-\frac{(4+k)}{5-k} \cdot -\frac{3}{2} = -1$
 $3(4+k) = -2(5-k)$
 $12 + 3k = -10 + 2k$
 $k = -22$
 $\therefore (4x + 5y - 2) - 22(x - y + 1) = 0$
 $4x + 5y - 2 - 22x + 22y - 22 = 0$
 $-18x + 27y - 24 = 0$
 $6x - 9y + 8 = 0$

Question 3

a) i) $3 \sin \theta = -2 \cos \theta$

$$\tan \theta = -\frac{2}{3}$$

2nd or 4th quadrant

$$\theta = 180 + 33^\circ 41' \text{ or}$$

$$360 - 33^\circ 41'$$

$$\theta = 146^\circ 19' \text{ or } 326^\circ 19'$$

ii) $\cos 2\theta = \cos \theta$

$$2\cos^2 \theta - 1 = \cos \theta$$

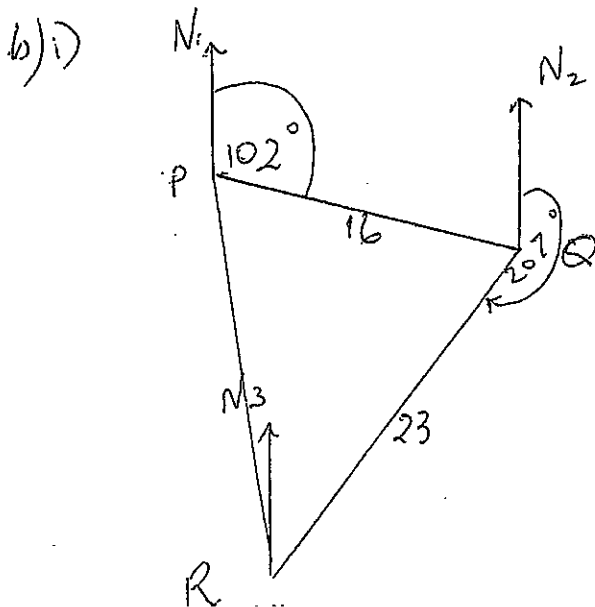
$$2\cos^2 \theta - \cos \theta - 1 = 0$$

$$(2\cos \theta + 1)(\cos \theta - 1) = 0$$

$$\cos \theta = -\frac{1}{2} \text{ or } 1$$

$$\theta = 180 - 60, 180 + 60, 0, 360$$

$$\theta = 120^\circ, 240^\circ, 0^\circ, 360^\circ$$



ii) $\angle PQR = 78^\circ$ (co-interior $\angle$'s $PN_1 \parallel QN_2$)

$$\therefore \angle PQR = 360 - 207 - 78$$

$$= 75^\circ \text{ (Angle sum of revolution)}$$

iii) $q^2 = p^2 + r^2 - 2pr \cos Q$

$$q^2 = 23^2 + 16^2 - 2 \times 23 \times 16 \times \cos 75^\circ$$

$$= 594.509$$

$$q = 24.38 \text{ km}$$

iv) First find either $\angle P$ or $\angle R$ in $\triangle PQR$:

$$\frac{\sin P}{23} = \frac{\sin 75^\circ}{24.38}$$

$$\sin P = \frac{23 \sin 75^\circ}{24.38}$$

$$= 0.9125$$

$$\angle P = 65^\circ 41'$$

(Acute angle required as $23 < 24.38$)

$\therefore$ Bearing of Yacht from P is $167^\circ 41'$

$\therefore$ Bearing of P from Yacht is $180 + 167^\circ 41' = 347^\circ 41'$

c) $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$

$$\cos(60^\circ + 45^\circ) = \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}}$$

$$= \frac{1 - \sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{2} - \sqrt{6}}{4}$$

$$d) \cos(180-A) = -\cos A$$

$$\sin(180+A) = -\sin A$$

$$\sin(-A) = -\sin A$$

$$\sin(90-A) = \cos A$$

$$\therefore LHS =$$

$$\frac{-\cos A}{-\sin A} - \frac{-\sin A}{\cos A}$$

$$= \cot A + \tan A$$

$$= \frac{1}{\tan A} + \tan A$$

$$= \frac{1 + \tan^2 A}{\tan A}$$

$$= \frac{\sec^2 A}{\tan A}$$

$$= \frac{1}{\cos^2 A} \div \tan A$$

$$= \frac{1}{\cos^2 A} \times \frac{\cos A}{\sin A}$$

$$= \frac{1}{\cos A} \times \frac{1}{\sin A}$$

$$= \sec A \operatorname{cosec} A$$